

Annika Rodriguez,
The University of Texas at San Antonio, San Antonio TX, 78249

Abstract

This study applies machine learning to predict concentrations of 35 PFAS target substances. Leveraging 26,900 observations, we identified geographic and atmospheric drivers of "Forever Chemical" seepage into the California water tab.

Introduction (Background)

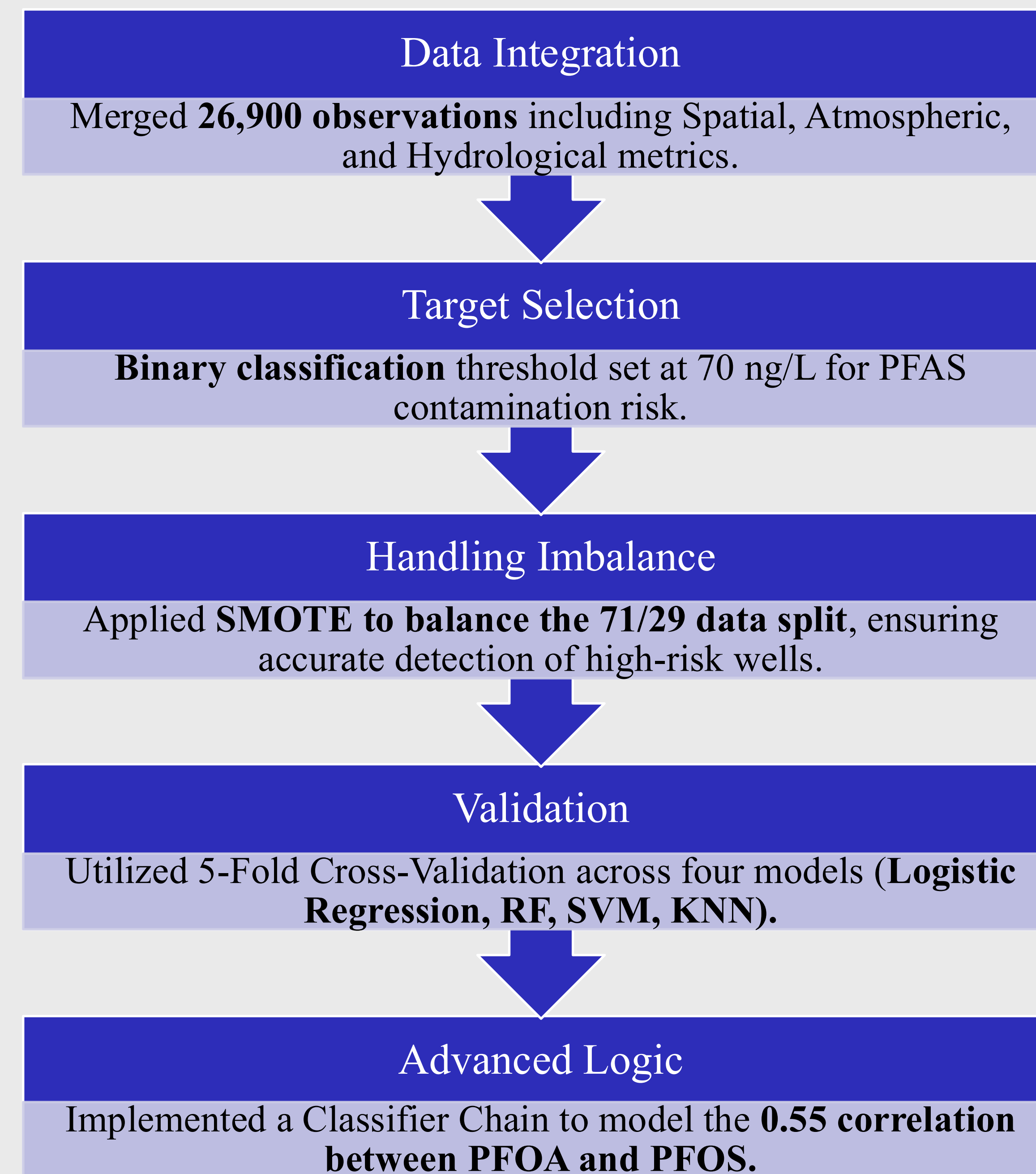
PFAS compounds are resistant to degradation due to carbon-fluorine bonds. Exposure is linked to severe health risks. Monitoring costs remain a barrier for public health officials.

- **The PFAS Crisis:** Per- and polyfluoroalkyl substances (PFAS) are industrial pollutants with carbon-fluorine bonds that resist natural degradation.
- **Public Health:** Exposure is linked to severe metabolic and immune system disorders. New 2026 regulations require stricter monitoring of individual chemicals like PFOA and PFOS.
- **The Data Challenge:** Identifying contamination is expensive and labor-intensive. Furthermore, the available data is highly imbalanced (71% clean vs. 29% contaminated), which often leads to "Majority Class Bias" in standard statistical models.

Purpose (Objective, Aim, Goal)

The primary goal of this study is to develop a scalable, cost-effective screening tool that identifies high-risk groundwater wells (>70 ng/L) using environmental and atmospheric predictors. We aim to overcome the statistical challenges of data imbalance and linear modeling limitations to provide high-sensitivity detection of "Forever Chemical" hotspots. Ultimately, this framework serves as a strategic roadmap for the California Water Board to prioritize laboratory resources and protect public health.

Methods



Results

- **Model Performance:** The baseline **Logistic Regression** failed significantly (0% sensitivity), proving that simple linear models cannot detect contamination in this dataset.
- **The Winner: Random Forest** emerged as the "Gold Standard," achieving an **AUROC of 0.9818** and a **sensitivity of ~93%**.
- **Reliability:** Calibration curves confirmed that Random Forest probability scores aligned most closely with real-world contamination frequencies.
- **Top Predictors:** Variable importance analysis identified **Latitude, and Longitude**, as the most influential factors in predicting PFAS presence.

Results - con't

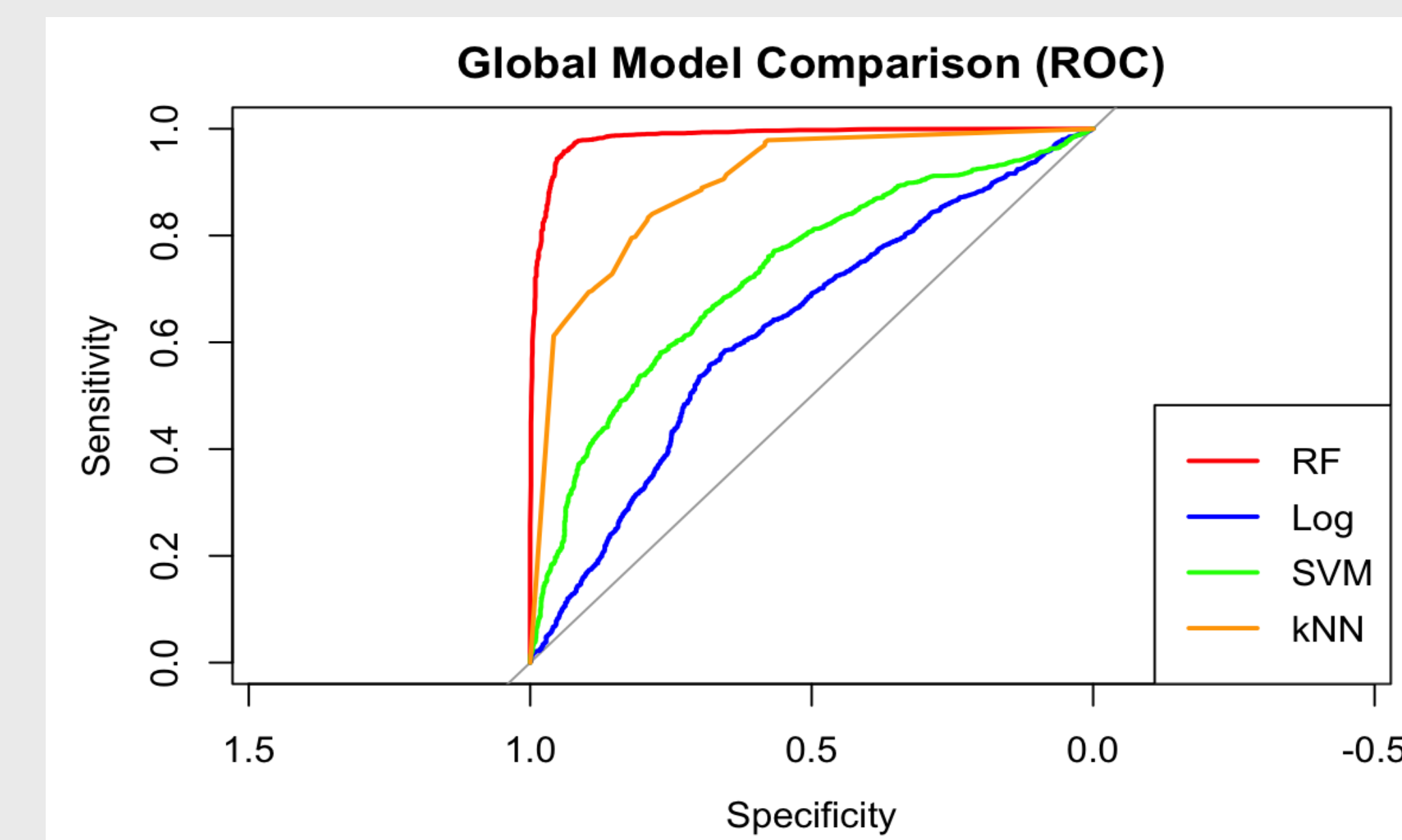


Figure 1: Model Performance Comparison. This plot shows the Area Under the ROC Curve. The Random Forest Curve displays a near perfect ability to distinguish between contaminated and clean wells compared to the other models.

Figure 2: Chemical & Environmental Correlation.

The dark blue tile between PFOA and PFOS represents a high 0.55 correlation. This statistical link justifies our use of a Classifier Chain, as the presence of one chemical is a strong predictor of the other.

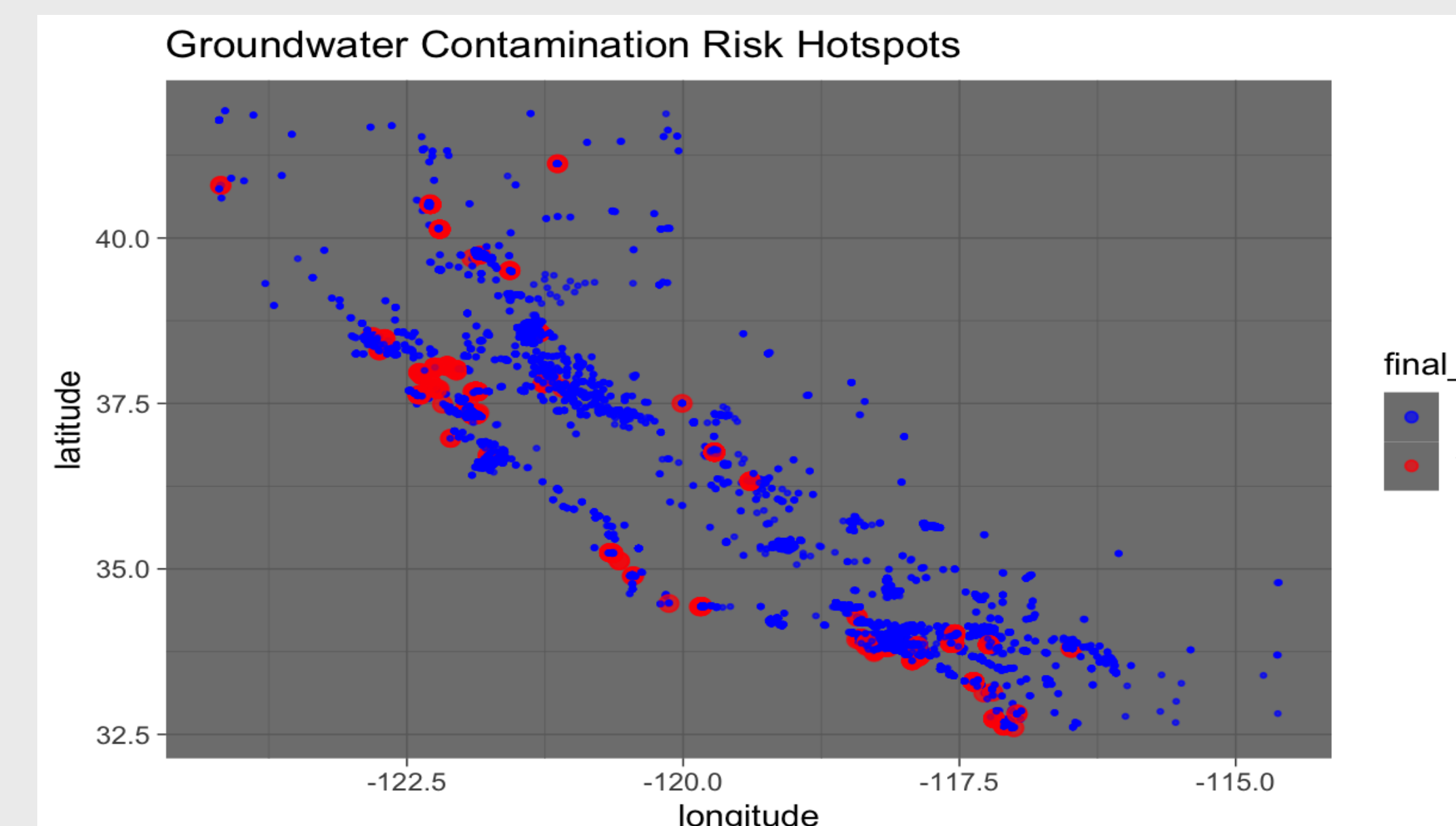
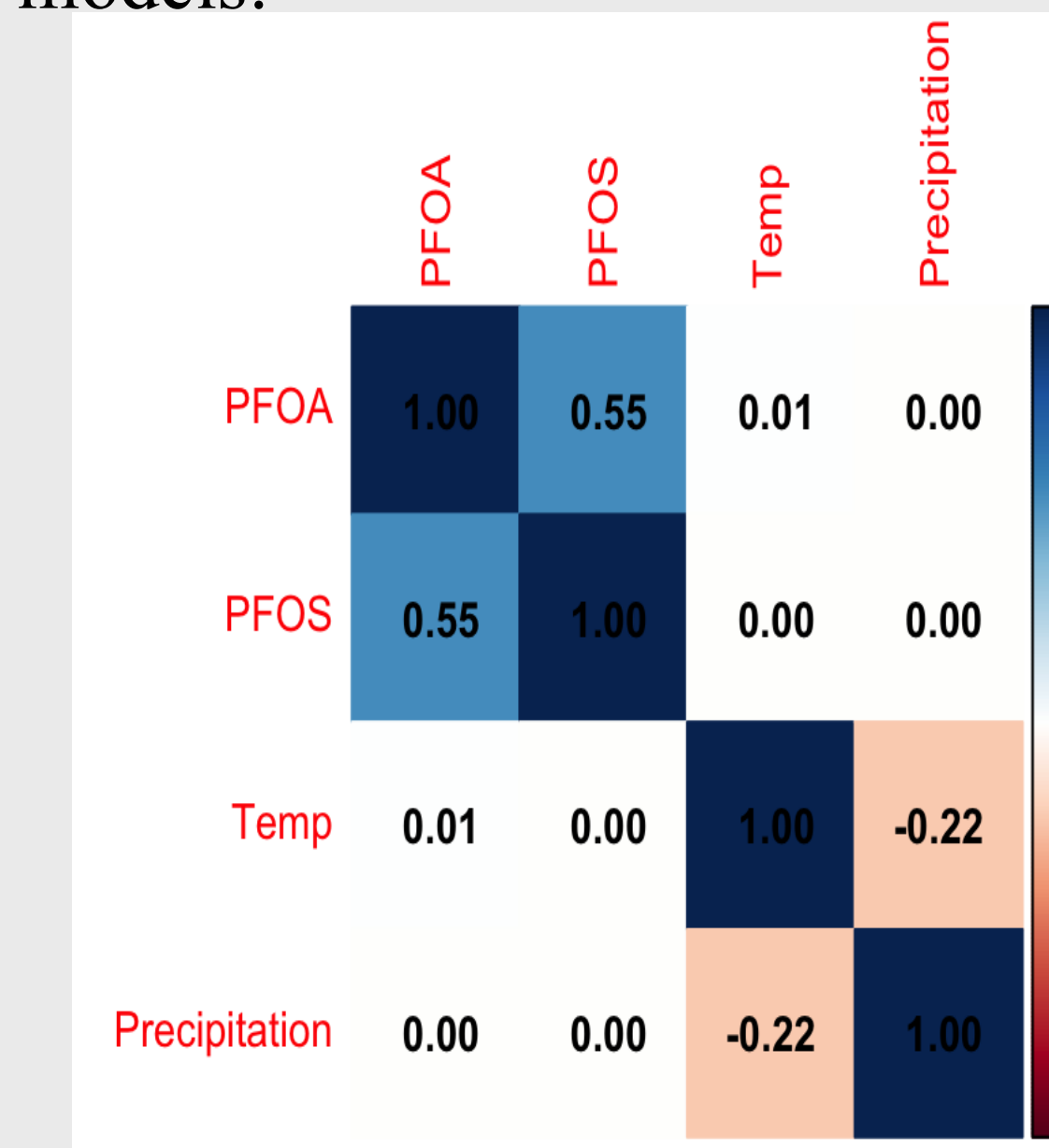


Figure 3: Predicted Groundwater Risk Hotspots. This map visualizes our model's final output across California. **Red points** indicate high-risk zones (>70 ng/L) where the Water Board should prioritize physical testing, while **blue points** indicate predicted safe zones.

Conclusions (Disc. or Summary)

➤ Summary of Results

- **Hypothesis Validation:** Non-linear models outperformed linear baselines, proving PFAS seepage is driven by complex environmental interactions.
- **Top Performer:** Random Forest 98.18% AUROC & ~93% sensitivity.
- **Chain Logic:** Classifier Chain successfully leveraged the 0.55 correlation between PFOA and PFOS to refine prediction accuracy.

➤ Real World Importance

- **Cost Efficiency:** Provides a low-cost screening tool for the California Water Board to prioritize expensive physical testing in high-risk "Red Zones."
- **Public Health:** Supports 2026 regulatory shifts by identifying 92% of contaminated sites using accessible environmental data.

➤ Future Directions:

- **Industrial Proximity:** Integrate distances to manufacturing sites to improve localized plume detection.
- **Full 35 Target Chain:** Expand the multi-label architecture to include all 35 target PFAS substances.
- **Hydrological Depth:** Incorporate groundwater flow direction and soil permeability model vertical chemical migration.

References

Liu et al. (2023). Prediction of 35 Target Per- and Polyfluoroalkyl Substances (PFASs) in California Groundwater Using Multilabel Semisupervised Machine Learning.
James, G., et al. (2021). An Introduction to Statistical Learning with Applications in R.

Acknowledgements

We would like to thank the UTSA Data Analytics program for providing the computational resources and guidance for this study.